

drought recovery 50yr summary

Drought recovery analysis (including 50-year)

Date: August 2026

Script: redo_recovery_with_50yr.py

Notebook: drought_recovery_comparison.ipynb
(DROUGHT_LENGTHS = [1, 3, 10, 50])

Figures: figures/

Sequence: 40 yr average spinup -> drought (dry) -> 5 yr recovery (average). Recovery plots align t=0 to drought end.

Baseline for anomalies

Domain	Member	Notes
Wolf	baseline (95 yr)	Same paths as short_baseline for years 0-54
Potomac	short_baseline (55 yr) + end-year snapshots	Long baseline raw exists through year 94 but 219 h is incomplete after year 70. Spatial maps at years 89/90/94 use cached snapshots under domains/potomac2/.../baseline/snapshots/. Recovery timeseries remap onto short_baseline years 50-54 (same intra-year phase) with a storage offset so year-1 end matches the year-90 snapshot. An earlier off-by-one phase slip made Potomac 50-yr dS look jagged; that is fixed.

Domain-total temporary vs persistent storage deficit

Temporary = recovered in recovery year 1; persistent = still missing after year 1 (106 m3).

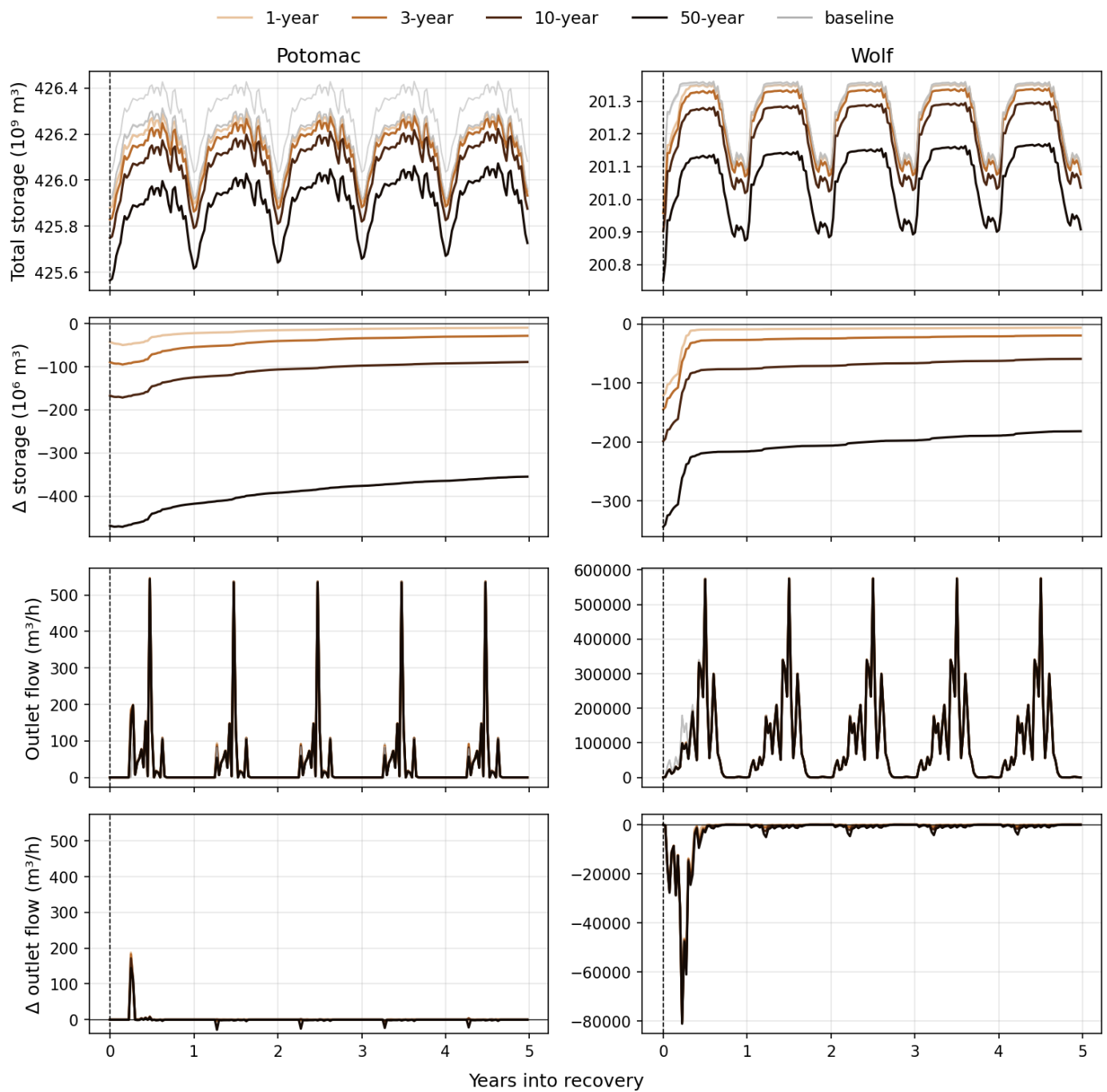
Domain	1-yr temp / persist	3-yr	10-yr	50-yr
Potomac	66 / 23	81 / 55	88 / 126	92 / 418
Wolf	117 / 9	123 / 27	127 / 76	133 / 216

Temporary saturates by ~10 years. Persistent keeps growing through 50 years (especially Potomac).

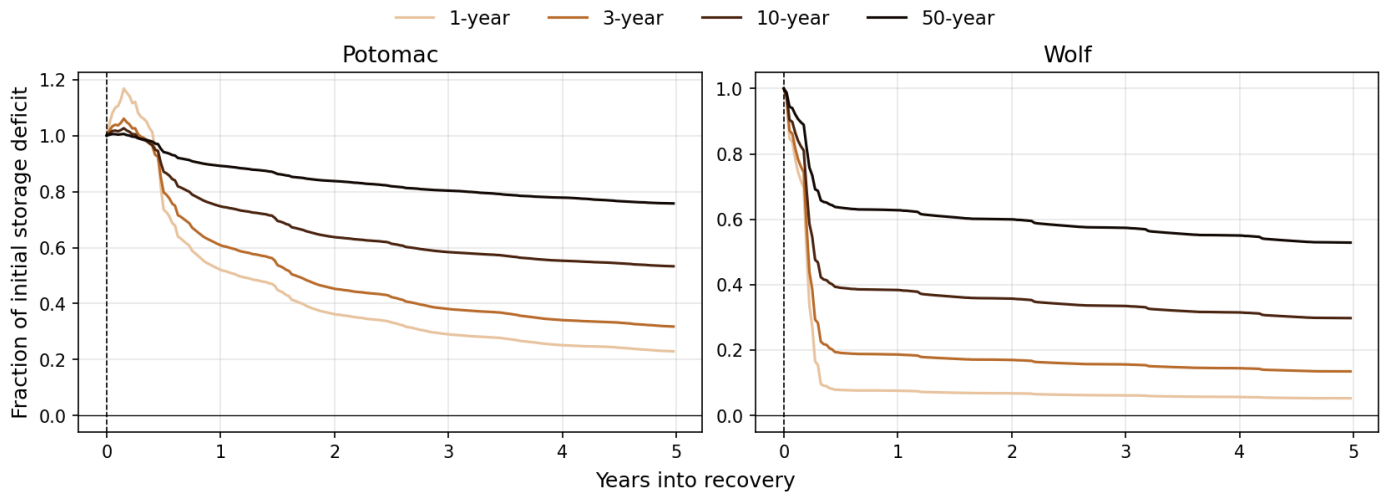
Figures

Recovery timeseries (by length)

- drought_recovery_totals_and_anomalies.png - storage, dS, outlet flow, dQ over 5-year recovery (1/3/10/50)

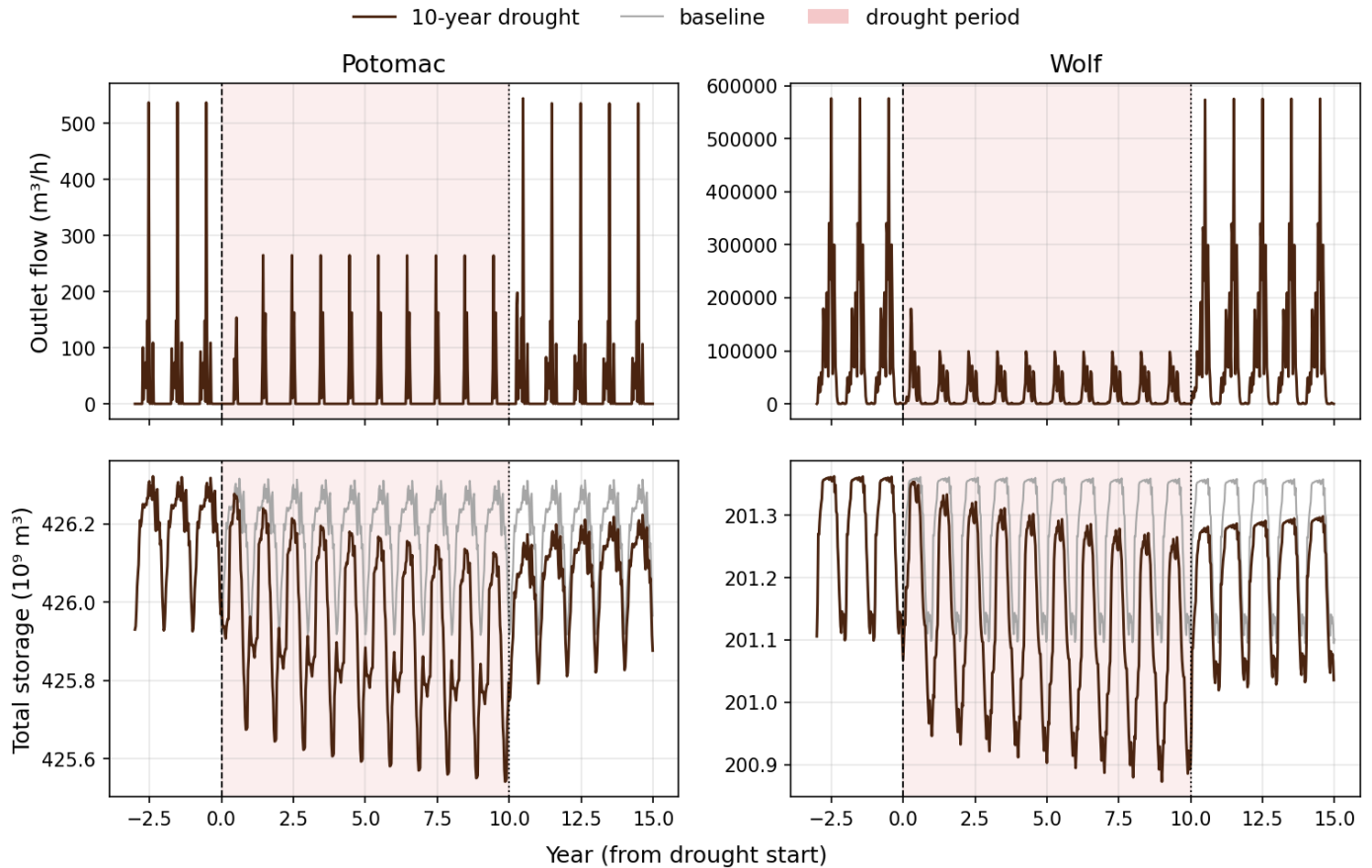


- drought_recovery_by_length_fractional_storage.png - fraction of initial storage deficit remaining

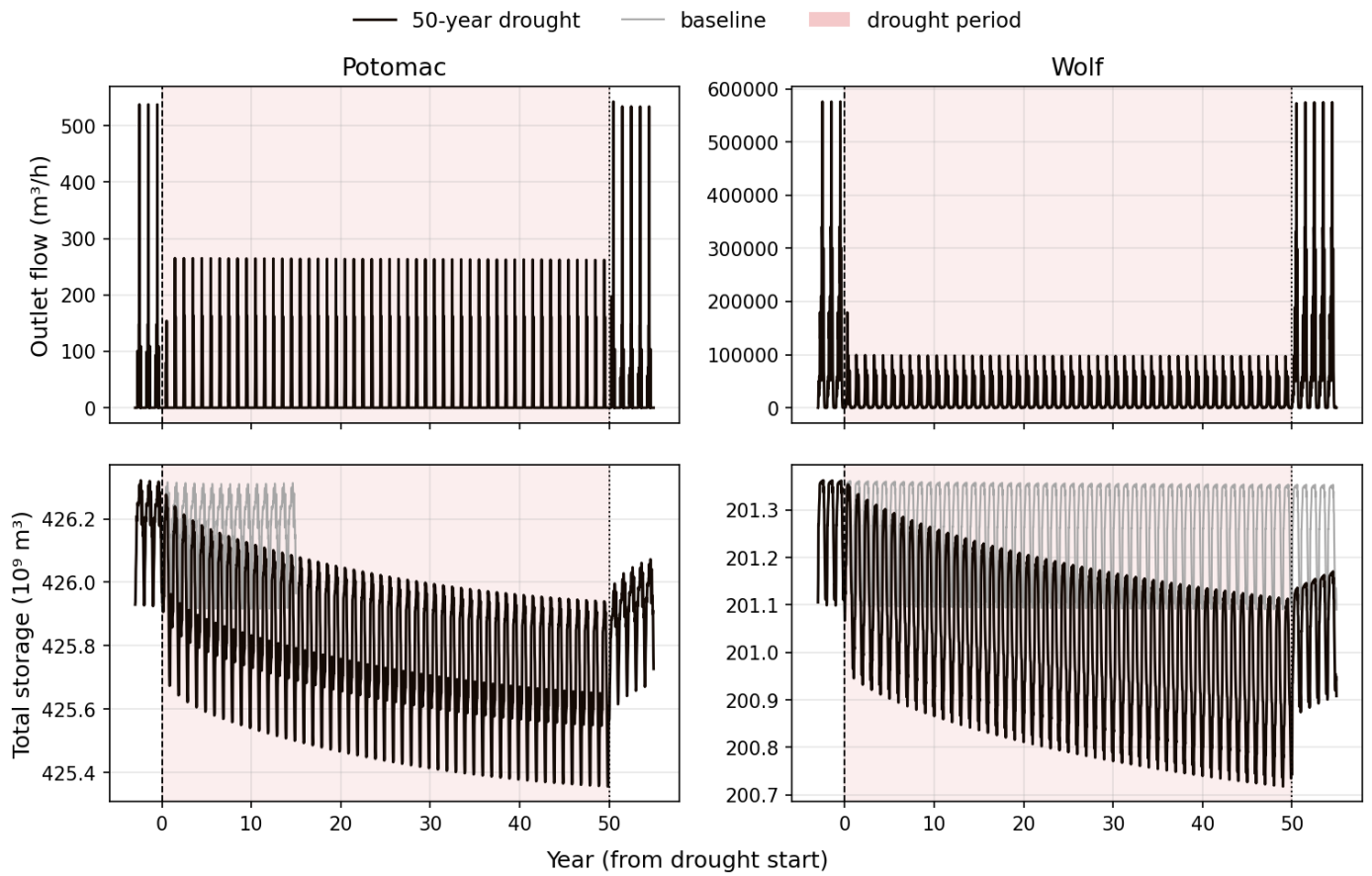


Drought course (+ recovery)

- ten_year_drought_streamflow_storage.png - last 3 spinup years -> 10-year drought -> recovery

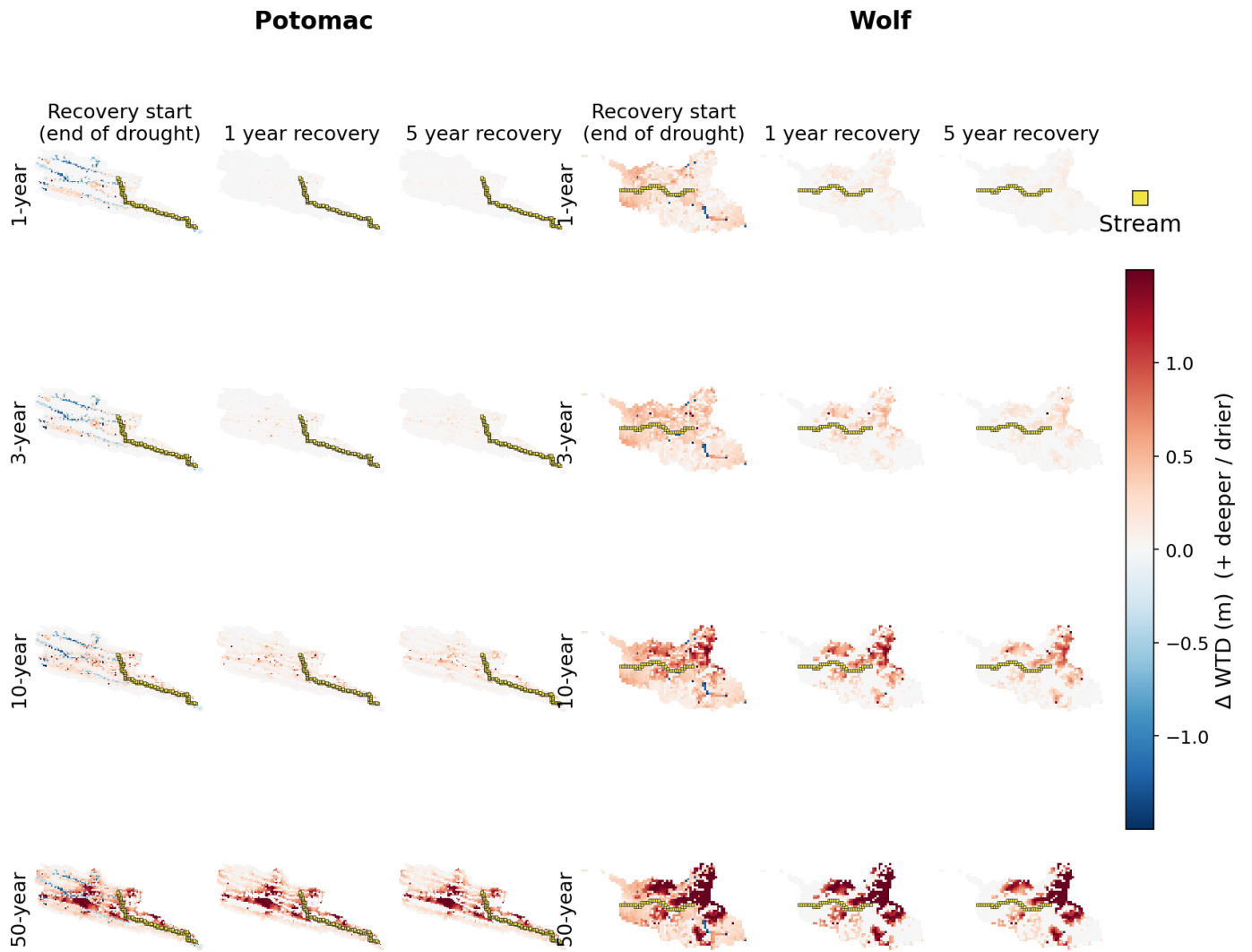


- fifty_year_drought_streamflow_storage.png - same layout for 50-year (new)



Spatial WTD / storage deficits

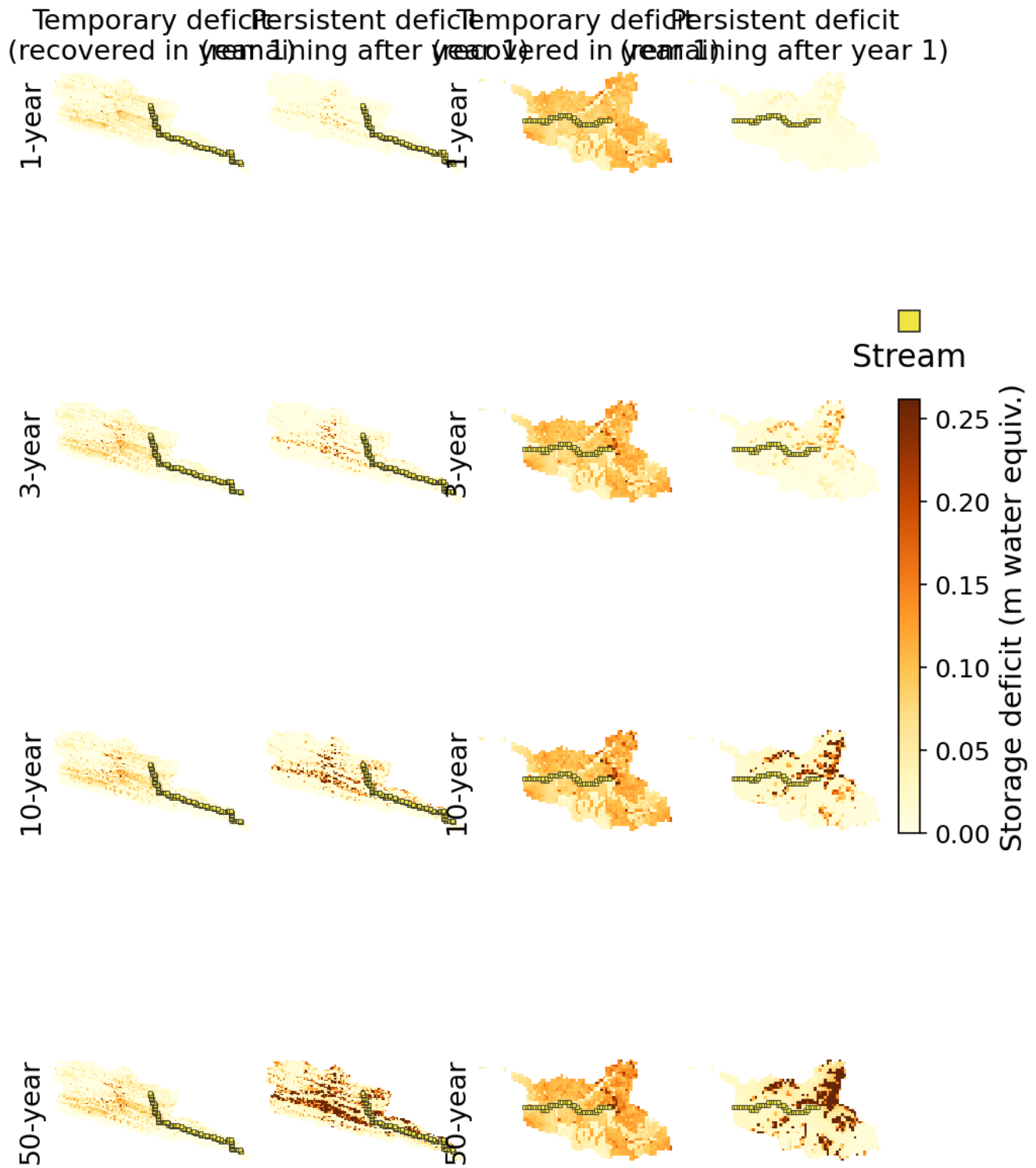
- drought_recovery_wtd_anomaly_maps.png - dWTD at recovery start, +1 yr, +5 yr (rows = 1/3/10/50)



- drought_recovery_temp_persistent_storage_deficit_maps.png - temporary vs persistent storage (m water equiv.)

Potomac

Wolf

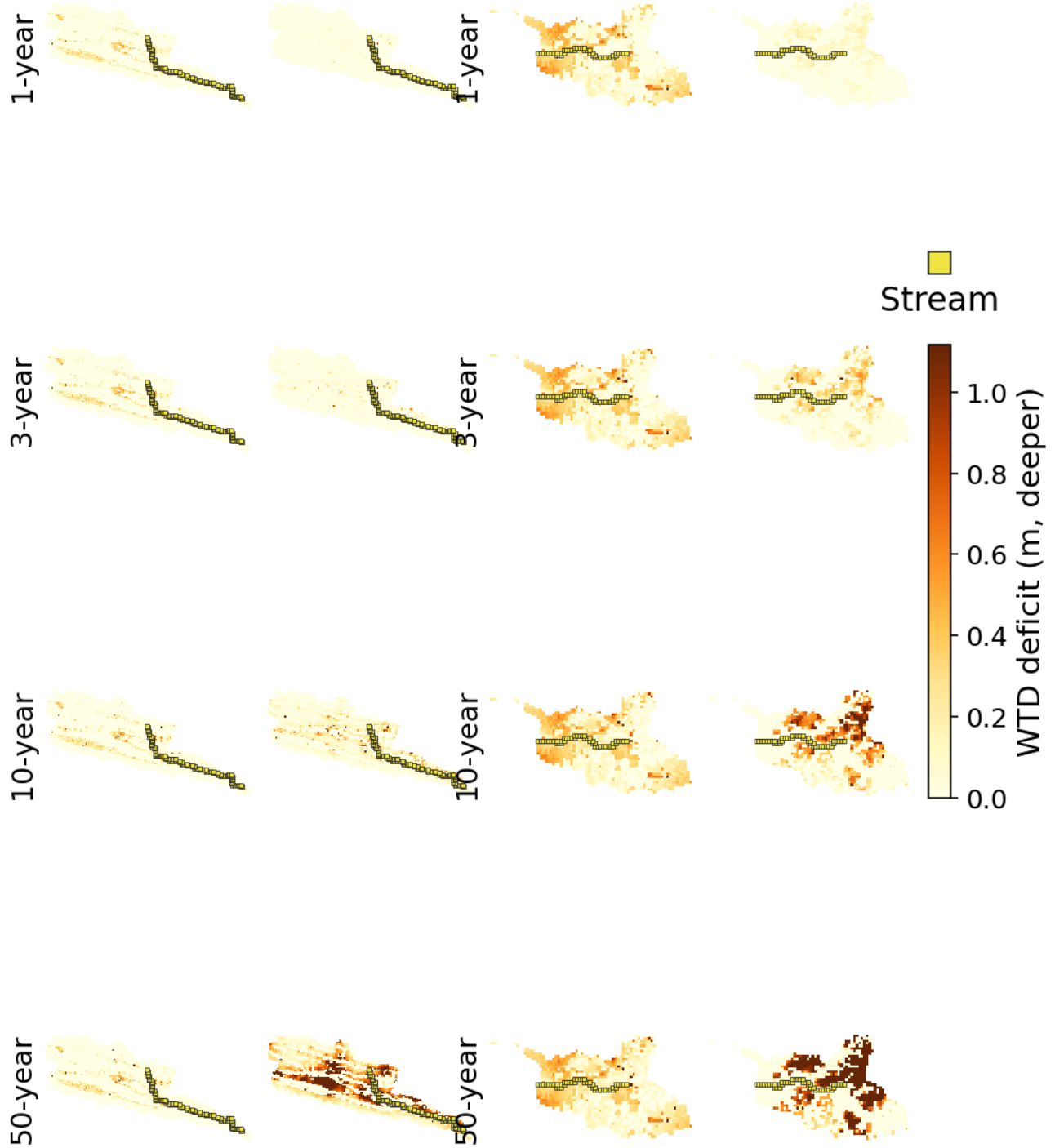


- drought_recovery_temp_persistent_wtd_deficit_maps.png - same split for dWTD

Potomac

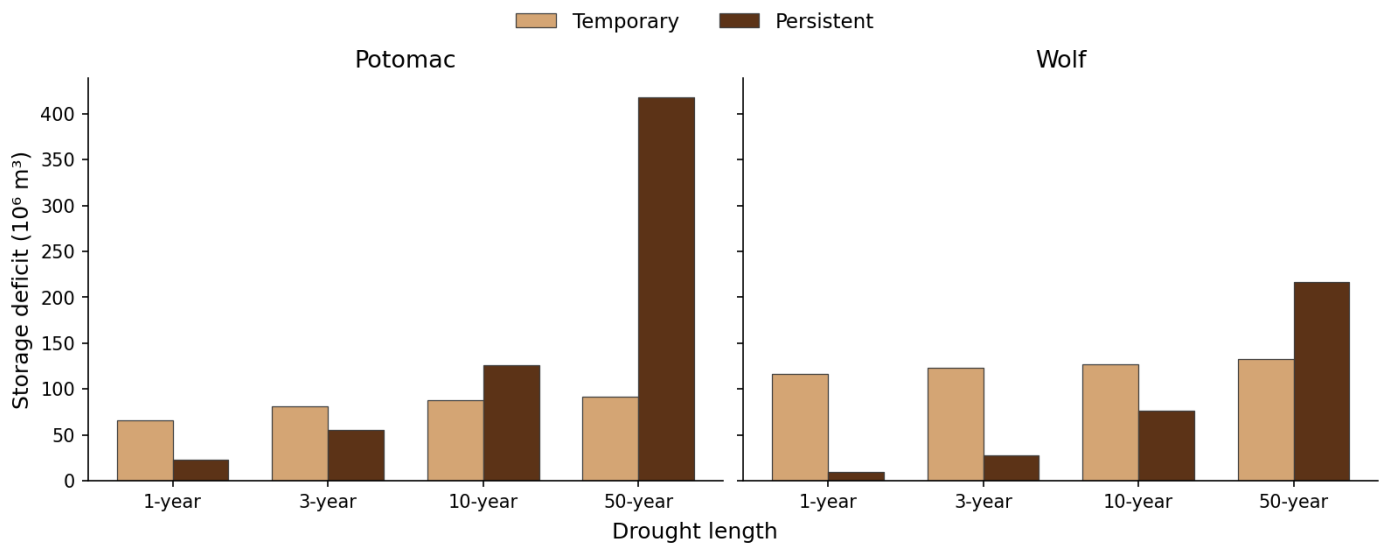
Wolf

Temporary deficit (recovered in year 1) Persistent deficit (remaining after year 1) Temporary deficit (recovered in year 1) Persistent deficit (remaining after year 1)

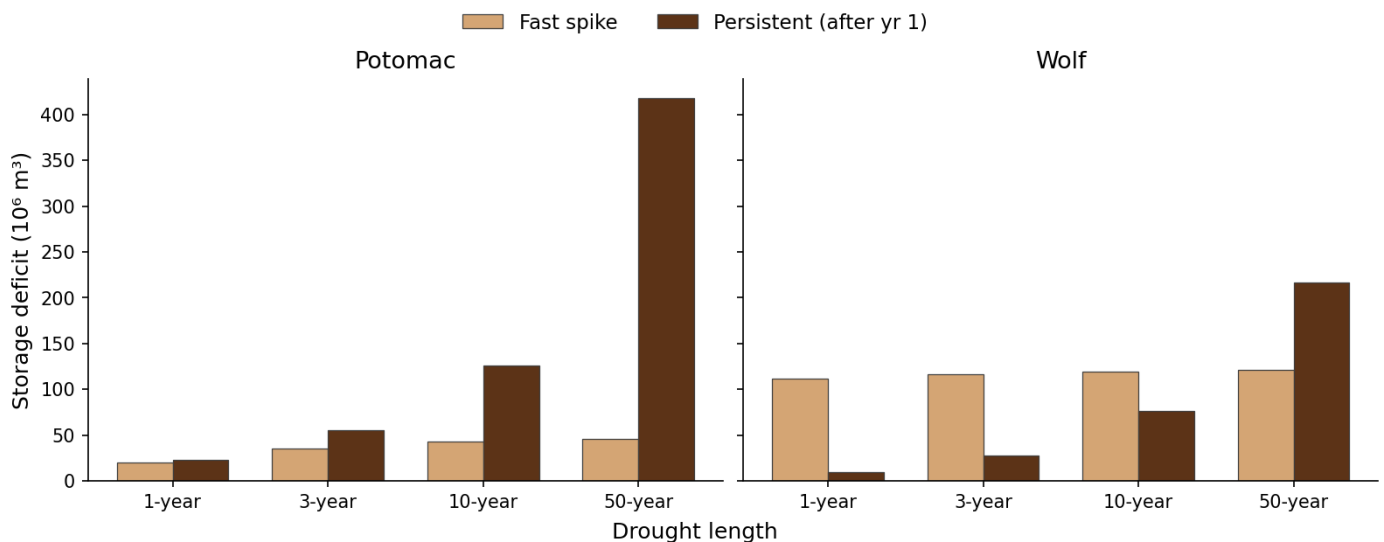


Totals by drought length

- drought_recovery_temp_persistent_deficit_bars.png - map-based domain totals



- drought_recovery_temp_persistent_deficit_bars_spike.png - fast-spike vs persistent (slow branch projected to $t=0$)



- [drought_recovery_temp_spike_correction.csv](figures/drought_recovery_temp_spike_correction.csv)

- breakpoint fits / spike magnitudes

Machine-readable totals

- [drought_recovery_50yr_totals.json](figures/drought_recovery_50yr_totals.json)

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Takeaways with 50-year included

1. Two-phase recovery still holds - fast year-1 recovery vs slow residual; the fast component barely grows from 10 -> 50 years.

2. Persistent deficit is where length matters - Potomac persistent $\sim 3\times$ the 10-year value at 50 years (126 $\rightarrow$ 418); Wolf $\sim 3\times$ as well (76 $\rightarrow$ 216).
3. Potomac accumulates more persistent deficit than Wolf at long durations, despite Wolf having a larger temporary (near-stream) pulse.
4. 5 years of average forcing is not enough to erase the 50-year memory in either domain (dS remains large at recovery year 5).

How to reproduce

```
```bash
conda activate /glade/work/bwest/conda-envs/droughts
source ~/pf_env.sh
cd /glade/derecho/scratch/bwest/drought-ensemble
python analysis/redo_recovery_with_50yr.py
```
```

Covariate / drainage / multivar screens from the earlier 10-year-focused follow-ups were not re-run here; say if you want those extended to 50-year as well.